

2.Data Science(Week 5-7) - Set -2

Total points 13/15

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✓ What is a characteristic feature of a normal distribution? * 1/1

- ☒ A) Symmetrical bell-shaped curve ✓
- ☐ B) Long right tail
- ☐ C) Flat shape with no peaks

✓ If the correlation coefficient between two variables is 0.9, what does it imply? *1/1

- ☒ A) Strong positive correlation ✓
- ☐ B) Weak correlation
- ☐ C) No correlation

✓ What does a positively skewed distribution indicate? * 1/1

- ☒ A) Data has a longer tail on the right side ✓
- ☐ B) Data has a longer tail on the left side
- ☐ C) Data is symmetrically distributed



✓ **When two variables move in opposite directions, what is the correlation called?** *1/1

☐ A) Positive correlation

☒ B) Negative correlation ✓

☐ C) Neutral correlation

✓ **What is the primary goal of Univariate Analysis?** * 1/1

☐ A) To study the relationship between two variables

☒ B) To summarize and describe the main characteristics of one variable ✓

☐ C) To predict future data points

✓ **Which measure is used to find how spread out data points are?** * 1/1

☐ A) Correlation

☒ B) Variance ✓

☐ C) Mode

✓ **Which measure is often used to detect outliers in univariate analysis?** * 1/1

☐ A) Mean

☒ B) Interquartile Range (IQR) ✓

☐ C) Correlation Coefficient

✓ In Bivariate Analysis, what is the main focus? *

1/1

- ☐ A) Analyzing one variable at a time
- ☒ B) Exploring the relationship between two variables ✓
- ☐ C) Creating machine learning models

✓ What is the primary purpose of using a heatmap in data visualization? *

1/1

- ☐ A) To compare time-series data
- ☒ B) To visualize the correlation between multiple variables ✓
- ☐ C) To show the median value of a dataset

✗ Which visualization is best for identifying the spread of a single variable?

*0/1

- ☐ A) Scatter Plot
- ☐ B) Histogram
- ☒ C) Line Plot ✗

Correct answer

- ☒ B) Histogram



✓ Which visualization is commonly used to show the frequency distribution of categorical data?

*1/1

☐ A) Line Chart

☒ B) Bar Plot



☐ C) Violin Plot

✗ Which central tendency measure is influenced most by outliers? *

0/1

☐ A) Mean

☒ B) Median



☐ C) Mode

Correct answer

☒ A) Mean

✓ Which plot is most suitable for analyzing the relationship between two continuous variables?

*1/1

☐ A) Box Plot

☒ B) Scatter Plot



☐ C) Pie Chart



✓ **What does a violin plot combine? ***

1/1

☒ A) Box plot and KDE (Kernel Density Estimate) ✓

☐ B) Histogram and Scatter Plot

☐ C) Pie Chart and Line Plot

✓ **In a box plot, what represents the median of the data? ***

1/1

☐ A) Top whisker

☐ B) Bottom whisker

☒ C) Central line inside the box ✓

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